Raphael Ferreira, PhD

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Bibliography overview

- 32 research articles
- 14 as first/co-first author (including *Science*, *Nature*, *PNAS*, *Nature Biotechnology*)
- 3 patent applications
- Citations: 2600+
- H-index: 20

Positions

Assistant Professor (Tenure Track) — DTU Health Tech
Aug 2025–present

Postdoctoral Fellow — Harvard University
Oct 2019–Jul 2025
Advisor: George Church

PhD Student — Chalmers University of Technology
2015–2019
Advisor: Jens Nielsen

Visiting PhD Student — Massachusetts Institute of Technology
2018
Advisor: Matthew Vander Heiden

Research Intern — Genethon
2014
Advisor: Federico Mingozzi

Research Intern — NASA
2013–2014
Advisor: Lynn J. Rothschild

Education

PhD, Bioscience (Systems Biology) — Chalmers University of Technology
2019

M.Sc., Synthetic Biology and Systems Biology — University of Paris
2014

Key publications

* *indicate co-first authors.*

1 Abbott, K. L.*, Subudhi, S.*, **Ferreira, R.***, Gültekin, Yş., Steinbuch, S. C., Munim, M. B., Ramesh, D. L., Honeder, S. E., Kumar, A. S., Wu, M. *et al.*, 2026. Nutrient requirements of organ-specific metastasis in breast cancer. *Nature.*

2 Perrotta, R. M.*, Vinke, S.*, **Ferreira, R***, Moret, M., Mahas, A., Chiappino-Pepe, A., Riedmayr, L. M., Mehra, A.-T., Lehmann, L. S. and Church, G. M., 2025. Engineered base editors with reduced bystander editing through directed evolution. *Nature Biotechnology* pp.1–10.

3 Koeppel, J.*, **Ferreira, R.***, Vanderstichele, T., Riedmayr, L. M., Peets, E. M., Girling, G., Weller, J., Murat, P., Liberante, F. G., Ellis, T. *et al.*, 2025. Randomizing the human genome by engineering recombination between repeat elements. *Science*, 387(6733), p.eado3979.

4 **Ferreira, R.***, Teixeira, P. G.*, Siewers, V. and Nielsen, J., 2018. Redirection of lipid flux toward phospholipids in yeast increases fatty acid turnover and secretion. *Proceedings of the National Academy of Sciences*, 115(6), pp.1262–1267.

Funding

Main applicant

- 2026 – 11M DKK: Lundbeck Foundation Fellow
- 2025 – 6.2M DKK: Danmarks Frie Forskningsfond Sapere Aude
- 2025 – 4.6M DKK: Novo Nordisk Foundation Start Package
- 2020 – 1.2M DKK: Knut and Alice Wallenberg Foundation – 2-year postdoctoral scholarship at Harvard
- 2018 – 65k DKK: Barbro Osher Pro Suecia Foundation – 3-month PhD scholarship at MIT

(Total raised = 23M DKK / approx. 3.1M EUR)

Contributor

- 2022 – 504k DKK: Wyss Institute validation project (Contributor)
- 2021 – 388k DKK: Q-FASTR Harvard Medical School (Contributor)
- 2020 – 197k DKK: NIH RADx for COVID-19 (Contributor)

Awards

- 2025 – Nominated to Sigma Xi Scientific Research Honor Society.
- 2023 – Fellow of 50 Fifty VC (one of 50 inspiring entrepreneurial scientists).

Teaching and supervision

PhD student:

1. Victoria Kalla (2026 –)
2. Anny Mais (2026 –)

Teaching

- 2025 – *From idea to project plan in biotech and pharmaceutical research* (DTU 22231; Guest lecturer)

Previous supervision and teaching (as PhD student / postdoc):

- Supervision: 6 Master's students, 1 B.Sc. student, and ~30 iGEM students.
- Teaching: *How To Grow Almost Anything* (TA, Harvard/MIT, 2023); *Molecular Biotechnology* (Lecturer, Chalmers, 2020–2021); *iGEM Chalmers team* (Supervisor, 2016, 2017, 2018); *Bioreaction Engineering* (TA, Chalmers, 2015)

Publications

* *indicate co-first authors.*

1 Riedmayr, L. M., Koeppel, J., Church, G. M., Parts, L. and **Ferreira, R.**, 2026. Genome structure interrogation via recombinase shuffling of multiplexed prime edits. *Nature Protocols* (accepted).

2 Mahas, A.*, **Ferreira, R.***, Riedmayr, L. M. and Church, G. M., 2026. Programmable nucleic acid sensing in human cells using circularizable ssDNA. *Nature Communications* (accepted).

3 Abbott, K. L.*, Subudhi, S.*, **Ferreira, R.***, Gültekin, Yş., Steinbuch, S. C., Munim, M. B., Ramesh, D. L., Honeder, S. E., Kumar, A. S., Wu, M. *et al.*, 2026. Nutrient requirements of organ-specific metastasis in breast cancer. *Nature*, pp.1–10.

4 Mahas, A., Perrotta, R. M., Oliynyk, R. T., **Ferreira, R.** and Church, G. M., 2026. Short single-stranded DNA oligonucleotides enable intracellular transcription of functional RNAs in mammalian cells. *bioRxiv*, 2026.05.30.729002.

5 Perrotta, R. M.*, Vinke, S.*, **Ferreira, R***, Moret, M., Mahas, A., Chiappino-Pepe, A., Riedmayr, L. M., Mehra, A.-T., Lehmann, L. S. and Church, G. M., 2025. Engineered base editors with reduced bystander editing through directed evolution. *Nature Biotechnology*, pp.1–10.

6 Koeppel, J.*, **Ferreira, R.***, Vanderstichele, T., Riedmayr, L. M., Peets, E. M., Girling, G., Weller, J., Murat, P., Liberante, F. G., Ellis, T. *et al.*, 2025. Randomizing the human genome by engineering recombination between repeat elements. *Science*, 387(6733), pp.eado3979.

7 Do, B. T.*, Hsu, P. P.*, Vermeulen, S. Y., Wang, Z., Hirz, T., Abbott, K. L., Aziz, N., Replogle, J. M., Bjelosevic, S., Paolino, J. *et al.*, 2024. Nucleotide depletion promotes cell fate transitions by inducing DNA replication stress. *Developmental Cell*, 59(16), pp.2203–2221.

8 Riedmayr, L. M., Hinrichsmeyer, K. S., Thalhammer, S. B., Mittas, D. M., Karguth, N., Otify, D. Y., Böhm, S., Weber, V. J., Bartoschek, M. D., Splith, V. *et al.*, 2023. mRNA trans-splicing dual AAV vectors for (epi) genome editing and gene therapy. *Nature Communications*, 14(1), p.6578.

9 Abbott, K. L.*, Ali, A.*, Casalena, D., Do, B. T., **Ferreira, R.**, Cheah, J. H., Soule, C. K., Deik, A., Kunchok, T., Schmidt, D. R. *et al.*, 2023. Screening in serum-derived medium reveals differential response to compounds targeting metabolism. *Cell Chemical Biology*, 30(9), pp.1156–1168.

10 **Ferreira, R.***, Hedin, K. A.*, Nielsen, J. and David, F., 2023. Biosensor-assisted laboratory evolution of malonyl-CoA production in *Saccharomyces cerevisiae*. *bioRxiv*.

11 Skrekas, C., **Ferreira, R.** and David, F., 2022. Fluorescence-activated cell sorting as a tool for recombinant strain screening. In: *Yeast Metabolic Engineering: Methods and Protocols*. Springer, pp.39–57.

12 Robinson-McCarthy, L. R.*, Mijalis, A. J.*, Filsinger, G. T., De Puig, H., Donghia, N. M., Schaus, T. E., Rasmussen, R. A., **Ferreira, R.**, Lunshof, J. E., Chao, G. *et al.*, 2021. Laboratory-generated DNA can cause anomalous pathogen diagnostic test results. *Microbiology Spectrum*, 9(2), pp.e00313–21.

13 Bergenholm, D.*, Dabirian, Y.*, **Ferreira, R.***, Siewers, V., David, F. and Nielsen, J., 2021. Rational gRNA design based on transcription factor binding data. *Synthetic Biology*, 6(1), p.ysab014.

14 Ferraro, G. B.*, Ali, A.*, Luengo, A.*, Kodack, D. P., Deik, A., Abbott, K. L., Bezwada, D., Blanc, L., Prideaux, B., Jin, X. *et al.*, 2021. Fatty acid synthesis is required for breast cancer brain metastasis. *Nature Cancer*, 2(4), pp.414–428.

15 Luengo, A., Li, Z., Gui, D. Y., Sullivan, L. B., Zagorulya, M., Do, B. T., **Ferreira, R.**, Naamati, A., Ali, A., Lewis, C. A. *et al.*, 2021. Increased demand for NAD^+ relative to ATP drives aerobic glycolysis. *Molecular Cell*, 81(4), pp.691–707.

16 Robinson-McCarthy, L. R., Mijalis, A. J., Filsinger, G. T., De Puig, H., Donghia, N. M., Schaus, T. E., Rasmussen, R. A., **Ferreira, R.**, Lunshof, J. E., Chao, G. *et al.*, 2021. Anomalous COVID-19 tests hinder researchers. *Science*, 371(6526), pp.244–245.

17 Smith, C. J.*, Castanon, O.*, Said, K., Volf, V., Khoshakhlagh, P., Hornick, A., **Ferreira, R.**, Wu, C.-T., Güell, M., Garg, S. *et al.*, 2020. Enabling large-scale genome editing at repetitive elements by reducing DNA nicking. *Nucleic Acids Research*, 48(9), pp.5183–5195.

18 Robinson, J. L.*, Kocabaş, Pınar*, Wang, H.*, Cholley, P.-E.*, Cook, D., Nilsson, A., Anton, M., **Ferreira, R.**, Domenzain, I., Billa, V. *et al.*, 2020. An atlas of human metabolism. *Science Signaling*, 13(624), pp.eaaz1482.

19 Lau, A. N., Li, Z., Danai, L. V., Westermark, A. M., Darnell, A. M., **Ferreira, R.**, Gocheva, V., Sivanand, S., Lien, E. C., Sapp, K. M. *et al.*, 2020. Dissecting cell-type-specific metabolism in pancreatic ductal adenocarcinoma. *eLife*, 9, pp.e56782.

20 Castanon, O.*, Smith, C. J.*, Khoshakhlagh, P., **Ferreira, R.**, Güell, M., Said, K., Yildiz, R., Dysart, M., Wang, S., Thompson, D. *et al.*, 2020. CRISPR-mediated biocontainment. *bioRxiv*.

21 Gatto, F., **Ferreira, R.** and Nielsen, J., 2020. Pan-cancer analysis of the metabolic reaction network. *Metabolic Engineering*, 57, pp.51–62.

22 **Ferreira, R.**, Skrekas, C., Hedin, A., Sánchez, B. J., Siewers, V., Nielsen, J. and David, F., 2019. Model-assisted fine-tuning of central carbon metabolism in yeast through dCas9-based regulation. *ACS Synthetic Biology*, 8(11), pp.2457–2463.

23 **Ferreira, R.**, Limeta, A. and Nielsen, J., 2019. Tackling cancer with yeast-based technologies. *Trends in Biotechnology*, 37(6), pp.592–603.

24 **Ferreira, R.**, Skrekas, C., Nielsen, J. and David, F., 2018. Multiplexed CRISPR/Cas9 genome editing and gene regulation using Csy4 in *Saccharomyces cerevisiae*. *ACS Synthetic Biology*, 7(1), pp.10–15.

25 **Ferreira, R.**, David, F. and Nielsen, J., 2018. Advancing biotechnology with CRISPR/Cas9: recent applications and patent landscape. *Journal of Industrial Microbiology and Biotechnology*, 45(7), pp.467–480.

26 **Ferreira, R.**, Teixeira, P. G., Gossing, M., David, F., Siewers, V. and Nielsen, J., 2018. Metabolic engineering of *Saccharomyces cerevisiae* for overproduction of triacylglycerols. *Metabolic Engineering Communications*, 6, pp.22–27.

27 **Ferreira, R.***, Teixeira, P. G.*, Siewers, V. and Nielsen, J., 2018. Redirection of lipid flux toward

phospholipids in yeast increases fatty acid turnover and secretion. *Proceedings of the National Academy of Sciences*, 115(6), pp.1262–1267.

28 Tippmann, S., **Ferreira, R.**, Siewers, V., Nielsen, J. and Chen, Y., 2017. Effects of acetoacetyl-CoA synthase expression on production of farnesene in *Saccharomyces cerevisiae*. *Journal of Industrial Microbiology and Biotechnology*, 44(6), pp.911–922.

29 Jensen, E. D.*, **Ferreira, R.***, Jakočiūnas, T., Arsovska, D., Zhang, J., Ding, L., Smith, J. D., David, F., Nielsen, J., Jensen, M. K. *et al.*, 2017. Transcriptional reprogramming in yeast using dCas9 and combinatorial gRNA strategies. *Microbial Cell Factories*, 16(1), p.46.

30 Teixeira, P. G., **Ferreira, R.**, Zhou, Y. J., Siewers, V. and Nielsen, J., 2017. Dynamic regulation of fatty acid pools for improved production of fatty alcohols in *Saccharomyces cerevisiae*. *Microbial Cell Factories*, 16(1), p.45.

31 **Ferreira, R.***, Gatto, F.* and Nielsen, J., 2017. Exploiting off-targeting in guide-RNA s for CRISPR systems for simultaneous editing of multiple genes. *FEBS Letters*, 591(20), pp.3288–3295.

32 Fujishima, K., Venter, C., Wang, K., **Ferreira, R.** and Rothschild, L. J., 2015. An overhang-based DNA block shuffling method for creating a customized random library. *Scientific Reports*, 5(1), p.9740.

Patents

- 2025 – COMPOSITIONS AND METHODS FOR PROGRAMMABLE NUCLEIC ACID SENSING – No. 63/760,371
- 2024 – METHODS FOR MODIFYING A CELL GENOME – PCT WO-2024170866-A1
- 2024 – ANTIGEN-SPECIFIC FUSION BY ENGINEERED CELLS – No. 63/725,961

Conference

- 2024 – Cold Spring Harbor CRISPR 2024 (New York, USA): *Randomizing the human genome via recombination between repeat elements* (Poster)
- 2023 – Cold Spring Harbor CRISPR 2023 (New York, USA): *Randomizing the human genome via recombination between repeat elements* (Talk)
- 2022 – Wyss Institute Retreat 2022 (Boston, USA): *Targeted cell-fusion for cancer therapy* (Poster)
- 2018 – AACR Special Conference "Convergence: Artificial Intelligence, Big Data and Prediction in Cancer" (Newport, USA): *Global analysis of gene expression changes associated with hypermutation* (Poster)
- 2018 – Winter Q-Bio (Hawaii, USA): *Multiplexed fine-tuning of central carbon metabolism using CRISPR/Cas gRNA libraries and a Malonyl-CoA biosensor* (Poster)
- 2016 – Metabolic Engineering XI (Kobe, Japan): *CRISPR-Cas assisted metabolic engineering of lipid metabolism in Saccharomyces cerevisiae* (Rapid-fire talk + Poster)

Other

Peer review: Nucleic Acids Research; Biotechnology Advances; ACS Synthetic Biology; FEMS Yeast Research; Scientific Reports.